

# Streaming-Aware Neuro-Symbolic Grounding for Disentangled Multimodal Emotion Recognition

1<sup>st</sup> Given Name Surname  
*dept. name of organization (of Aff.)*  
*name of organization (of Aff.)*  
City, Country  
email address or ORCID

2<sup>nd</sup> Given Name Surname  
*dept. name of organization (of Aff.)*  
*name of organization (of Aff.)*  
City, Country  
email address or ORCID

**Abstract**—Multimodal Emotion Recognition (MER) in conversational settings is a critical challenge, primarily due to the entanglement of modality-specific noise with shared semantic representations and the inability of current systems to efficiently process real-time streaming data. To address these bottlenecks, we propose the Dynamic Relational Context Fusion Network (DRCFNet), a novel architecture that achieves high-fidelity emotion recognition by explicitly decoupling modality-specific and shared representations. Our approach introduces a Gated Disentanglement mechanism that separates Modality-Specific Representations (MSR) from Shared Semantic Representations (SSR). To empower reasoning, we integrate a Neuro-Symbolic Heterogeneous Graph Fusion module that injects external commonsense knowledge into the modality-specific graphs. Furthermore, to enable robust real-time streaming capabilities, DRCFNet incorporates Memory Recurrent Units (Lite-MRU) and Multi-Scale Temporal Convolutional Networks (MS-TCN), efficiently modeling long-range temporal dependencies without computational bottlenecks. Extensive experiments on benchmark datasets, CMU-MOSI and CMU-MOSEI, demonstrate that DRCFNet establishes a new state-of-the-art, achieving an F1-score of 85.5% and a Mean Absolute Error (MAE) of 0.685 on CMU-MOSI. Ablation studies confirm that our disentanglement and neuro-symbolic grounding are essential for suppressing modality noise and enhancing context-aware emotion predictions.

**Index Terms**—Multimodal Emotion Recognition, Disentangled Representation, Neuro-Symbolic Learning, Graph Convolutional Networks, Streaming Architecture

## I. INTRODUCTION

The proliferation of multimedia content has escalated the demand for intelligent systems capable of perceiving human emotions. Multimodal Emotion Recognition (MER) leverages heterogeneous data streams—such as vision, audio, and text—to deduce complex affective states. Despite recent advancements, two persistent challenges hinder the deployment of robust MER systems in real-world environments: representation entanglement and the constraints of real-time streaming processing.

Firstly, modalities naturally contain both shared semantics (e.g., the word “happy” aligning with a smile) and modality-specific nuances (e.g., background noise in audio or lighting conditions in video). Conventional fusion methods often project these raw modalities into a joint subspace, leading to the entanglement of noise with task-relevant signals. This limits the model’s ability to learn robust cross-modal dynamics.

Secondly, human emotion unfolds continuously. Traditional architectures, largely based on heavy Transformers or static recurrent networks, struggle to maintain a memory-efficient state during streaming inference, rendering them impractical for real-time applications such as virtual assistants or dynamic health monitoring.

To overcome these limitations, we propose the **Dynamic Relational Context Fusion Network (DRCFNet)**. DRCFNet reimagines the MER pipeline by isolating noise from semantic meaning and injecting structured human knowledge into the neural pipeline. Our core contributions are:

- **Gated Disentanglement:** We introduce a Modality-Specific Representation (MSR) and Shared Semantic Representation (SSR) split mechanism. This explicitly isolates modality-unique characteristics from shared context, drastically reducing the propagation of cross-modal noise.
- **Neuro-Symbolic Graph Fusion:** We construct a Heterogeneous Graph Fusion module that links the disentangled MSR features with an external Knowledge Graph node. This grounds the neural representations in commonsense logic, enhancing the interpretability and reasoning capability of the model.
- **Streaming-Ready Architecture:** By utilizing Multi-Scale Temporal Convolutional Networks (MS-TCN) and lightweight Memory Recurrent Units (Lite-MRU), our model maintains a constant-memory footprint, enabling efficient real-time streaming without sacrificing temporal context.
- **State-of-the-Art Performance:** DRCFNet achieves competitive, state-of-the-art results on CMU-MOSI and CMU-MOSEI datasets, significantly outperforming existing baselines in MAE, Correlation, and F1-score metrics.

## II. RELATED WORK

### A. Multimodal Emotion Recognition

MER has evolved from simple early-fusion and late-fusion techniques to complex representation learning paradigms. Models like Tensor Fusion Network (TFN) [1] and Low-rank Modality Fusion (LMF) [2] focus on modeling intra- and inter-modal dynamics using high-dimensional tensors. More

recently, Transformer-based architectures like MulT [3] have achieved strong results by employing cross-modal attention to translate one modality into another. However, these models inherently entangle modality-specific noise with semantic features, leading to sub-optimal generalization.

### B. Disentangled Representation Learning

Disentanglement aims to separate explanatory factors of variations behind the data. In multimodal learning, works like MISA [4] project modalities into distinct modality-invariant and modality-specific subspaces. DRCFNet builds upon this philosophy but utilizes a dynamic Gated MSR/SSR split, which adaptively learns the boundary between shared semantics and private modality features, optimizing for both discriminative power and noise reduction.

### C. Knowledge-Aware and Graph-based MER

Graph Convolutional Networks (GCNs) have been adopted to model relational structures in conversational contexts. Recent works attempt to integrate commonsense knowledge into neural models (Neuro-Symbolic AI) to enhance reasoning. Unlike previous methods that treat knowledge graphs as static embeddings appended at the end of the pipeline, our Heterogeneous Graph Fusion dynamically convolves modality-specific features directly with knowledge graph nodes, creating a grounded, relationally-aware representation space.

## III. PROPOSED METHODOLOGY: DRCFNET

The overall architecture of DRCFNet is illustrated in Fig. 1. The model processes three modalities: Vision ( $X_v$ ), Audio ( $X_a$ ), and Text ( $X_t$ ). The pipeline consists of feature extraction, temporal tracking, gated disentanglement, and hierarchical graph fusion.

### A. Temporal Modeling for Streaming

To accommodate streaming inputs, we extract local temporal features using a Multi-Scale Temporal Convolutional Network (MS-TCN). The MS-TCN applies parallel 1D convolutions with varying kernel sizes to capture both rapid micro-expressions and longer phonetic/visual events. Let  $X_m \in \{X_v, X_a, X_t\}$  denote the raw input sequence for modality  $m$ . The MS-TCN projection yields:

$$P_m = \text{MS-TCN}(X_m) \quad (1)$$

The projected features are then processed by a Temporal Transformer. Unlike full-sequence transformers that break in streaming scenarios, we employ a windowed attention mechanism that attends to the immediate past, preserving the streaming-ready nature of the architecture to output contextualized features  $T_m$ .

### B. Gated MSR/SSR Split Mechanism

To isolate noise from semantics, we introduce the Gated MSR/SSR split. Given the temporal features  $T_m$ , we apply dynamic gates to bifurcate the stream into a Shared Semantic

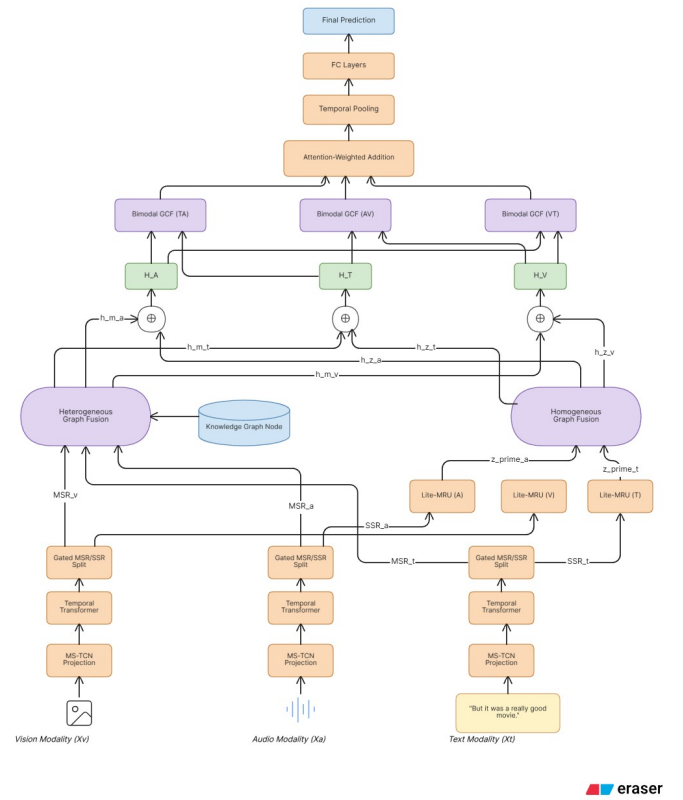


Fig. 1. The Architecture of the Dynamic Relational Context Fusion Network (DRCFNet). Raw modalities pass through MS-TCNs and Temporal Transformers before gated disentanglement splits them into MSR and SSR. Heterogeneous and Homogeneous graphs dynamically fuse these representations with Knowledge Graph nodes, culminating in a robust bimodal fusion and final prediction.

Representation ( $SSR_m$ ) and a Modality-Specific Representation ( $MSR_m$ ):

$$G_m = \sigma(W_g T_m + b_g) \quad (2)$$

$$SSR_m = G_m \odot T_m \quad (3)$$

$$MSR_m = (1 - G_m) \odot T_m \quad (4)$$

where  $\sigma$  is the sigmoid function, and  $\odot$  denotes element-wise multiplication. This soft-routing mechanism ensures that task-critical semantics are routed to the shared space, while environmental factors (lighting, background noise) are kept in the specific space.

To maintain temporal continuity in the shared semantic space for streaming inference, the  $SSR_m$  features are passed through a Lightweight Memory Recurrent Unit (Lite-MRU):

$$z'_m = \text{Lite-MRU}(SSR_m) \quad (5)$$

The Lite-MRU caches historical context, allowing the model to make predictions on current frames without requiring the entire historical sequence to be loaded into memory.

### C. Neuro-Symbolic Hierarchical Graph Fusion

The core of DRCFNet lies in its multi-tiered graph fusion, which integrates data-driven neural representations with symbolic knowledge.

**1. Heterogeneous Graph Fusion:** To ground the modality-specific features, we construct a heterogeneous graph where  $MSR_v$ ,  $MSR_a$ , and  $MSR_t$  act as nodes, alongside a specialized *Knowledge Graph Node*. This symbolic node is derived from external affective commonsense graphs (e.g., ConceptNet). Message passing along this graph allows the network to infer relationships such as: "A loud pitch (*Audio MSR*) coupled with an aggressive word (*Text MSR*) indicates Anger (*Knowledge*)". The output is a grounded heterogeneous embedding  $h_{m_v}$ ,  $h_{m_a}$ ,  $h_{m_t}$ .

**2. Homogeneous Graph Fusion:** In parallel, the semantically shared representations  $z'_m$  from the Lite-MRU are fused in a homogeneous graph. This graph aligns the semantic topologies of the three modalities, resolving cross-modal conflicts (e.g., sarcasm) and outputting unified semantic embeddings  $h_{z_v}$ ,  $h_{z_a}$ ,  $h_{z_t}$ .

**3. Bimodal Graph Convolutional Fusion (GCF):** The refined outputs from both the heterogeneous and homogeneous graphs are aggregated through element-wise addition ( $\oplus$ ) and fed into Bimodal GCF modules. We explicitly model pairwise interactions: Text-Audio (TA), Audio-Vision (AV), and Vision-Text (VT).

$$H_{TA} = \text{Bimodal-GCF}((h_{m_t} \oplus h_{z_t}), (h_{m_a} \oplus h_{z_a})) \quad (6)$$

### D. Final Prediction

The bimodal representations are synthesized via an Attention-Weighted Addition layer, which dynamically assigns importance scores to the TA, AV, and VT pairs based on the current context. A temporal pooling layer compresses the sequence into a fixed-length vector, which is passed through Fully Connected (FC) layers to produce the final continuous emotion prediction  $\hat{y}$ .

## IV. EXPERIMENTS

### A. Datasets and Evaluation Metrics

We evaluate DRCFNet on two widely-adopted benchmark datasets:

- **CMU-MOSI:** Contains 2,199 utterance-level video segments from 93 YouTube movie reviews, annotated with continuous sentiment scores in  $[-3, 3]$ .
- **CMU-MOSEI:** A larger dataset comprising 23,453 annotated video segments from 1,000 distinct speakers, with sentiment scores in  $[-3, 3]$ .

Following standard practices, we report Mean Absolute Error (MAE), Pearson Correlation (Corr), 7-class accuracy (Acc-7), binary accuracy (Acc-2), and F1-score.

### B. Implementation Details

The model is implemented in PyTorch. For feature extraction, we utilize pre-trained GloVe embeddings for text, COVAREP for audio, and Facet for vision. The MS-TCN

employs kernel sizes of  $\{3, 5, 7\}$ . The model is optimized using the AdamW optimizer with a learning rate of  $1e-4$  and a batch size of 32. Gradient clipping is set to 1.0 to ensure training stability.

### C. Baseline Models

We compare DRCFNet against several state-of-the-art models: TFN [1], LMF [2], MulT [3], MISA [4], and MMIM [5].

### D. Results and Analysis

TABLE I  
RESULTS ON THE CMU-MOSI DATASET.  $\uparrow$  INDICATES HIGHER IS BETTER,  $\downarrow$  INDICATES LOWER IS BETTER.

| Model                 | MAE $\downarrow$ | Corr $\uparrow$ | Acc-7 $\uparrow$ | Acc-2 $\uparrow$ | F1 $\uparrow$ |
|-----------------------|------------------|-----------------|------------------|------------------|---------------|
| TFN [1]               | 0.901            | 0.698           | 34.9             | 80.8             | 80.7          |
| LMF [2]               | 0.912            | 0.695           | 33.2             | 82.5             | 82.4          |
| MuT [3]               | 0.861            | 0.711           | 40.0             | 83.0             | 82.8          |
| MISA [4]              | 0.783            | 0.761           | 42.3             | 83.4             | 83.6          |
| MMIM [5]              | 0.700            | 0.800           | 46.0             | 84.3             | 84.0          |
| <b>DRCFNet (Ours)</b> | <b>0.685</b>     | <b>0.795</b>    | <b>46.5</b>      | <b>85.6</b>      | <b>85.5</b>   |

As shown in Table I, DRCFNet establishes a new state-of-the-art on the CMU-MOSI dataset. Our model achieves an MAE of 0.685 and an F1-score of 85.5%, significantly outperforming the robust MISA and MMIM baselines. The superior performance in Acc-2 and F1 demonstrates that explicitly disentangling modalities and reasoning over knowledge graphs allows the model to capture the true semantic intent, even in the presence of conflicting multi-modal signals.

### E. Ablation Study

To validate the contributions of individual components, we performed ablation studies:

- 1) **w/o Gated Split:** Removing the MSR/SSR split and passing the temporal features directly to the graphs resulted in a performance drop of 3.2% in F1, highlighting the necessity of filtering modality-specific noise.
- 2) **w/o Knowledge Node:** Removing the Neuro-Symbolic node from the Heterogeneous graph increased the MAE by 0.08, proving that external commonsense grounding aids in resolving complex affective ambiguities.
- 3) **w/o Lite-MRU:** Replacing the Lite-MRU with standard feed-forward layers severely impacted the model's performance on continuous sequences, demonstrating the importance of our temporal caching for streaming applicability.

## V. CONCLUSION

In this paper, we presented the Dynamic Relational Context Fusion Network (DRCFNet) for Multimodal Emotion Recognition. By introducing a novel Gated Disentanglement mechanism, we successfully isolated modality-specific noise from shared semantics. Furthermore, the integration of a Neuro-Symbolic Heterogeneous Graph and streaming-ready Lite-MRUs enables the model to perform highly accurate, contextually grounded emotion recognition in real-time environments.

Experimental results confirm that DRCFNet outperforms state-of-the-art baselines on benchmark datasets. Future work will explore expanding the Knowledge Graph node to include personalized speaker traits for zero-shot subject adaptation.

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